10.2 Classical Fields: the Legendre Transformation and Surface Terms

Two pieces of classical machinery are used from the first lectures on: the Legendre transformation, which takes us from a Lagrangian to a Hamiltonian, and the rule that surface terms may be dropped when we vary an action.

The Legendre Transformation

Definition and Mathematical Formulation

- **Single Variable**: For a convex function $f(x)$, the Legendre transform $g(p)$ is defined as:

$$g(p) = px(p) - f(x(p)), \quad \text{where} \quad p = \frac{df}{dx}.$$

  Here, $x(p)$ is obtained by inverting $p = df/dx$.
- **Multiple Variables**: For $f(x_1, \ldots, x_n)$, the transform for the first m variables is:

$$g(p_1, \ldots, p_m, x_{m+1}, \ldots, x_n) = \sum_{i=1}^{m} p_i x_i - f, \quad p_i = \frac{\partial f}{\partial x_i}.$$

  The untransformed variables $x_{m+1}, \ldots, x_n$ ride along as parameters.

Key Properties

- **Involutive**: Applying the transform twice returns f if f is convex.
- **Convexity Requirement**: Ensures uniqueness of $x(p)$. Non-convex functions require the Legendre–Fenchel transform.
- **Derivatives swap roles**: $dg/dp = x$, the inverse of $df/dx = p$. The slope of one function is the variable of the other.

Applications in Physics

Classical mechanics. The transition from the **Lagrangian** $L(q, \dot{q}, t)$ to the **Hamiltonian** $H(q, p, t)$:

$$p_i = \frac{\partial L}{\partial \dot{q}_i}, \quad H = \sum_i p_i \dot{q}_i - L.$$

This yields Hamilton's first-order equations.

Thermodynamics. Thermodynamic potentials are related via Legendre transforms:

$$H(S,p) = U + pV,$$
$$F(T,V) = U - TS,$$
$$G(T,p) = H - TS.$$

Each potential shifts control variables (e.g., $T \leftrightarrow S$). With the definition above, F and G are *minus* the Legendre transforms; the overall sign is a convention.

Field theory. From the **Lagrangian density** $\mathcal{L}(\phi, \partial_\mu \phi)$ to the **Hamiltonian density**:

$$\pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}, \quad \mathcal{H} = \pi\dot{\phi} - \mathcal{L}.$$

Geometric Interpretation

The Legendre transform constructs $g(p)$ as the maximum value of $px - f(x)$ for each p. This encodes $f(x)$ through its tangent lines (slopes p, intercepts $-g(p)$).

Examples

- Quadratic function: $f(x) = \frac{1}{2}kx^2 \to g(p) = \frac{p^2}{2k}$.
- Exponential function: $f(x) = e^x \to g(p) = p(\ln p - 1)$.

Advanced Topics

- **Symplectic Geometry**: Framework for Hamiltonian mechanics.
- **Contact Geometry**: Describes thermodynamic equilibrium spaces.
- **Hamilton–Jacobi Theory**: The generating function of the canonical transformation to constant variables obeys a PDE, the Hamilton–Jacobi equation.

In short, the Legendre transformation switches between conjugate variables, enabling reformulations like Hamiltonian mechanics and thermodynamic potentials. It preserves the information in the function under convexity, adapting the description to the natural variables of the problem.

Discarding Surface Terms in the Variational Principle

In quantum field theory, when deriving equations of motion such as the Klein–Gordon equation via the principle of least action, boundary terms arising from integration by parts are discarded under two justified assumptions:

1. Vanishing Field Variations at Boundaries: The variational principle considers small perturbations $\delta\phi(x)$ around the classical field configuration. These variations are required to have *compact support*, meaning $\delta\phi(x) \to 0$ at spacetime infinity.[1] For example, when varying the action $S = \int d^4x\, \mathcal{L}(\phi, \partial_\mu\phi)$, integration by parts generates boundary terms of the form

$$\int d^4x\, \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi \right) = \oint_{\text{boundary}} d\Sigma_\mu \, \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi,$$

which vanish because $\delta\phi = 0$ on the boundary. This ensures surface terms do not contribute to the equations of motion, regardless of the original field $\phi(x)$'s behavior. The vanishing of $\delta\phi$ at boundaries is a formal requirement of the variational principle, not a physical assumption about $\phi(x)$.

2. Physical Boundary Conditions for Fields: While the field $\phi(x)$ itself might oscillate rapidly at infinity, physical consistency demands fields behave "nicely" at spacetime boundaries. For quantum fields, this is often enforced by:

- *Square-integrability*: $\phi \in L^2$. This controls the field on average; it does not by itself make $\phi(x)$ vanish pointwise at infinity (§10.5).
- *Tempered distributions*: Fields are assumed to not blow up uncontrollably (standard in axiomatic QFT).
- *Schwartz functions*: Fields and derivatives decay rapidly, ensuring boundary terms involving $\phi(x)$ are well-controlled.

Heuristically, rapid oscillations of $\phi(x)$ are further suppressed in the path integral formalism. Configurations with wildly oscillating phases e^{iS} destructively interfere (the *stationary phase approximation*), leaving only contributions near classical paths, where the action is stationary. The Klein–Gordon equation arises as the condition for the action to be stationary under these variations. Crucially, boundary terms depend on $\delta\phi$ (not ϕ), and $\delta\phi$ vanishes at infinity by construction. Thus, discarding surface terms is valid in the variational framework, even if $\phi(x)$'s behavior at infinity appears ambiguous.

[1] The physics argument: we want the amplitude between given initial and final states, which are asymptotic states. With those specified, there are no functional variations at infinity.